

REVIEW ARTICLE



Standard use of Geographic Information System (GIS) techniques in honey bee research 2.0

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ABSTRACT

Geographic Information Systems (GIS) have been used in various fields and disciplines to summarize and analyse spatial patterns and distributions for the purpose of understanding how geographic and non-geographic entities interact with each other over space and time. Although honey bees are directly related to and influenced by their local environment, few studies have incorporated honey bee data into GIS for the purposes of gauging these spatial relationships. This paper will briefly discuss some of the types of spatial analyses and GIS methods that have been used for bees, and also, how some methodologies developed to study non-managed bees could be applied to honey bee research. With this paper, we aim to stimulate spatial thinking processes and thus the future use of GIS and spatial analyses to better understand the relationships between environmental characteristics and honey bee health and abundance. We will introduce the framework and some important basic concepts of GIS, as well as provide detailed instructions for becoming familiar and comfortable in using the GIS software ArcGIS Pro and Quantum GIS (QGIS) (commercial and free GIS packages, respectively) for the basics of geospatial research.

Uso estándar de las técnicas del Sistema de Información Geográfica (SIG) en la investigación de la abeja de la miel 2.0

Los Sistemas de Información Geográfica (SIG) se han utilizado en diversos campos y disciplinas para resumir y analizar patrones y distribuciones espaciales con el fin de comprender cómo interactúan entre sí entidades geográficas y no geográficas a lo largo del espacio y el tiempo. Aunque las abejas melíferas están directamente relacionadas con su entorno local e influidas por él, son pocos los estudios que han incorporado datos sobre ellas a los SIG con el fin de calibrar estas relaciones espaciales. En este artículo se expondrán brevemente algunos de los tipos de análisis espaciales y métodos SIG que se han utilizado para las abejas, y también cómo algunas metodologías desarrolladas para estudiar abejas no manejadas podrían aplicarse a la investigación de las abejas melíferas. Con esta contribución pretendemos estimular los procesos de reflexión espacial y, por ende, el uso futuro de los SIG y los análisis espaciales para comprender mejor las relaciones entre las características ambientales y la salud y abundancia de las abejas melíferas. Presentaremos el marco y algunos conceptos básicos importantes de los SIG, así como instrucciones detalladas para familiarizarse y sentirse cómodo en el uso de los programas informáticos SIG ArcGIS Pro y Quantum GIS (QGIS) (paquetes SIG comercial y gratuito, respectivamente) para los fundamentos de la investigación geoespacial.

在蜜蜂研究中规范使用地理信息系统 (GIS) 技术 2.0

地理信息系统 (GIS) 已被用于各个领域和学科, 用于总结 和分析空间模式和分布, 以了解地理实体和非地理实体在空间和时间上如何相互作用。

虽然蜜蜂与当地环境直接相关并受其影响, 但是很少有研究将蜜蜂数据纳入地理信息系统, 以衡量这些空间关系。本文将简要讨论一些用于蜜蜂的空间分析类型和地理信息系统方法, 以及如何把一些为研究非人工饲养的蜜蜂开发的方法应用于蜜蜂研究。本文旨在激发空间思维过程, 从而在未来利用地理信息系统和空间分析更好地理解环境特征与蜜蜂健康和数量之间的关系。我们将介绍地理信息系统的框架和一些重要的基本概念, 并详细说明如何熟悉和自如地使用地理信息系统软件 ArcGIS Pro 和 Quantum GIS (QGIS) (分别为商业和免费的地理信息系统软件 包) 进行地理空间研究的基础知识。

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1. Introduction

Due to the close relationship between honey bees and their surroundings, and because of recent increases in colony mortality in many regions of the world (Carreck & Neumann, 2010; Williams et al., 2010), there is an urgent need to better understand how environmental changes affect habitat, life patterns, and overall health of honey bees. This type of research can be facilitated through the use of Geographic Information Systems (GIS), which can briefly be described as a software system for handling and analysing map data through the use of coordinate systems and georeferenced information to aid in the analysis of spatial patterns that may exist between entities on the Earth's surface (Burrough & McDonnell, 1998). Because of its broad application, GIS methods are used in a diverse array of disciplines, including geoscience (Bonham-Carter, 1994), economics (Pogodzinski & Kos, 2012), and biology (Kozak et al., 2008). For example, GIS has been used extensively to undertake spatial analyses

of the effects of land-use and climate change on declines of non-managed bee abundance and health (Arthur et al., 2010; Biesmeijer et al., 2006; Chifflet et al., 2011; Choi et al., 2012; Fitzpatrick et al., 2007; Giannini et al., 2012; Kremen et al., 2002, 2004; Watson et al., 2011; Williams et al., 2012). Through the use of spatial analyses, researchers using GIS were able to determine that non-managed bees were more efficient crop pollinators when access to nearby forests was provided (Arthur et al., 2010; Kremen et al., 2004; Watson et al., 2011). Additionally, relationships between climate and non-managed bee species richness and abundance have been identified using GIS modelling procedures (Giannini et al., 2012). When this paper was initially written (Rogers & Staub, 2013), there were a limited number of honey bee studies that integrated GIS into their research (Berardinelli & Vedova, 2004; Henry et al., 2012; Naug, 2009). Since that time, there has been an increase in the use of GIS and spatial analyses for honey bee investigations which have increased visibility in the use of GIS for this

topic and uncovered new insights about the relationships between honey bees and their environment across space and time (e.g., Calovi et al., 2021; Douglas et al., 2020, 2022; Insolia et al., 2022; Jordan et al., 2021; Ochungo et al., 2021; Overturf et al., 2022; Quinlan et al., 2022, 2023; Rajagopalan et al., 2024; Richardson et al., 2023; Robinson et al., 2021; Switanek et al., 2017; Von Büren et al., 2019). Specifically, recent studies that have integrated spatial data sets to understand insect contributions to the economic value of crops across large geographic scales (Jordan et al., 2021), the spatial distribution of pesticide use and risk to bees (Douglas et al., 2020, 2022), foraging resources for bees (Ochungo et al., 2021), the contribution of land use, habitat quality, land management, weather and climate to honey production (Quinlan et al., 2022, 2023) and honey bee health (e.g., Calovi et al., 2021; Insolia et al., 2022; Overturf et al., 2022; Richardson et al., 2023). With this paper, we aim to continue to stimulate spatial thinking processes and thus the future use of GIS analyses to better understand the relationships between environmental characteristics and honey bee health and abundance for a broader geographic overview of the mechanisms affecting their plight (Naug, 2009). This paper introduces the framework and some important basic concepts of GIS and provides detailed instructions for using the GIS software ArcGIS Pro and Quantum GIS (QGIS) for the spatial analyses of honey bees. In this updated version of the paper, we have revised the original tutorials. We hope that the knowledge gained from this paper will further the field of honey bee science, but also promote the use of GIS in other fields of study.

2. What is GIS?

2.1. Definition

Geographical Information Systems (GIS) are computer systems based on hardware, software, and georeferenced data that can be used to collect, store, manage, process, analyse, and visualize both spatial and non-spatial information representing real-world geographic phenomena (Burrough & Mcdonnell,

1998; Neteler & Mitasova, 2008). Georeferenced data refers to any data that are linked to a location on the Earth's surface through the use of a geographic or projected coordinate system. GIS data are digital objects which represent real-world entities (Longley et al., 1999) and are defined by: their geometric properties (spatial location), their attributes (characteristics associated with each object), and their topology (definition of how entities are related to others in space) (Burrough & Mcdonnell, 1998). In other words, data provide means to locate them in space and can be overlaid, calculated, manipulated, visualised and analysed along with other data layers that use the same coordinate system. Each entity in the real world is represented by a data layer with geometric and topologic properties and an associated set of attributes, in the form of a table, which define the characteristics of that entity. GIS facilitates the analysis of spatial relationships within datasets based on the topological properties within the data. Topology refers to the interconnectivity and interrelated properties between data and defines and describes how spatial objects relate to their neighbours in space. This geographic descriptor is what sets GIS data apart from other data types (Burrough & Mcdonnell, 1998) (for a list of free GIS books and other information sources see Table 1).

2.2. GIS framework

The main principles of GIS (geometric properties, attributes, and topology) are built into two data structures: vector (e.g., points, lines, and polygons) and raster (e.g., climatic, altitude, or satellite imagery). The terms "layer" or "dataset" are used to denote a raster or a vector file type that contains a similar theme, for example, varying elevations across a terrain or locations of honey bee colonies, respectively. It is possible to convert a raster layer to a vector layer and vice versa, depending on what type of analysis will be conducted with the data. Each layer contains information about its geometric properties and a table of attributes associated to and linked

Table 1. Publication names, authors, and websites of free GIS Books and information sources.

Name of publication	Author(s), year	Website
Map Projections: A Working Manual	Snyder (1987)	http://pubs.er.usgs.gov/publication/pp1395
Geospatial Analysis – A comprehensive guide	De Smith et al. (2007)	http://www.spatialanalysisonline.com
A Practical Guide to Geostatistical Mapping	Hengl (2011)	http://www.lulu.com/shop/tomislav-hengl/a-practical-guide-to-geostatistical-mapping/ebook/product-17389015.html
GIS Commons: An Introductory Textbook on Geographic Information Systems	Schmandt (2009)	http://giscommons.org/
Various ESRI E-books	Many	http://www.esri.com/industries/ebooks
Essential of Geographic Information Systems	Campbell and Shin (2011)	https://open.umn.edu/opentextbooks/textbooks/essentials-of-geographic-information-systems
Instructional (and educational) Resources and the GIS & T Body of Knowledge	University Consortium for Geographic Information Science (UCGIS) (2022)	https://www.ucgis.org/instructional-resources https://www.ucgis.org/gis-t-body-of-knowledge

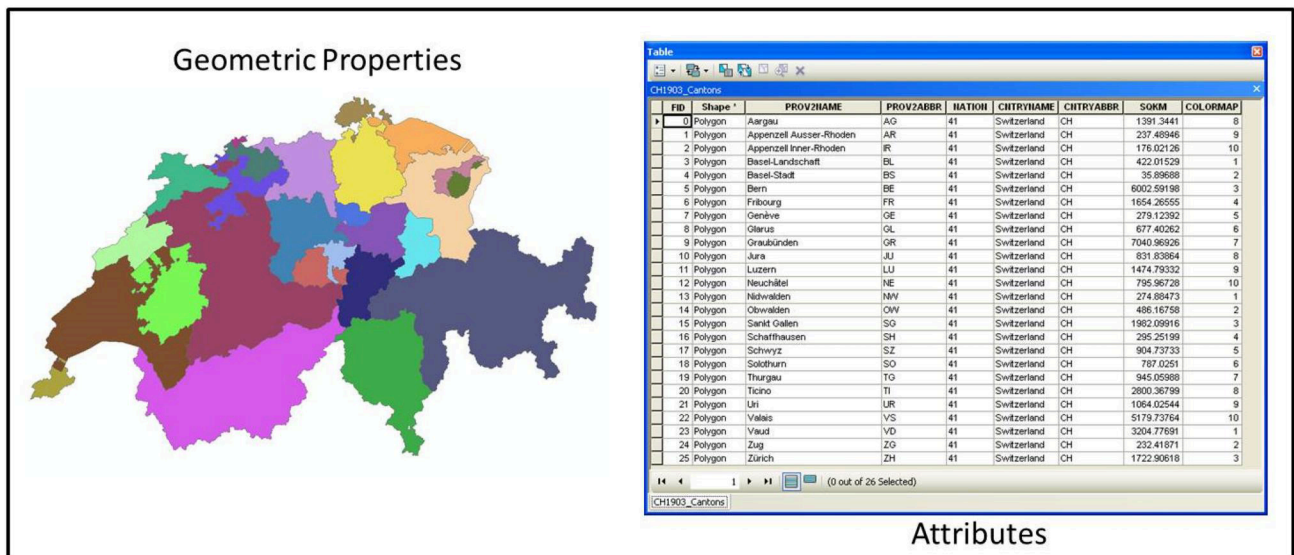


Figure 1. The GIS data framework is made up of both geometric properties and attributes associated with the geometry. A row in the attribute table exists for every polygon on the map. Some of the attributes for this Swiss layer include the names of cantons and their abbreviations, the country code, the country name and abbreviation, and the area. Reproduced with permission of Swisstopo (BA13016).

with, its respective geometric properties (Figure 1). You can view each layer's properties in GIS to see the resolution, file type, size of the file, and other information of importance.

2.2.1. Vector data structure

Vector layers represent discrete features in space in the forms of points, lines, and polygons. They represent unchanging static entities and do not contain spatial or temporal information (Burrough & Mcdonnell, 1998). For example, a polygon layer could represent the extent of an apiary, and a point layer could represent a honey bee colony on a map. Line layers can represent rivers, roads, tracks, and any other linear features. In Figure 2, a polygon, point, and line layer are shown as examples.

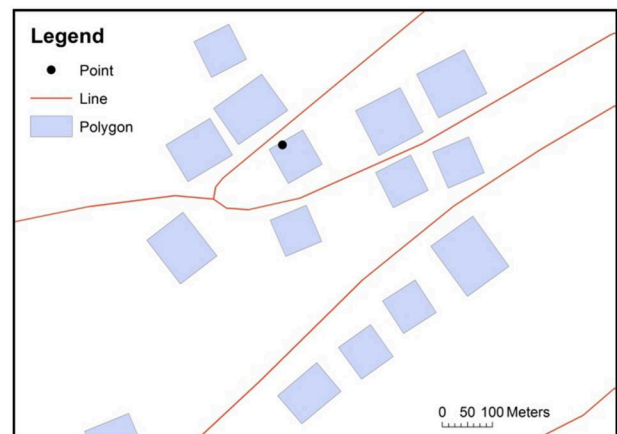


Figure 2. A map displaying a point, lines, and polygons. Reproduced with permission of Swisstopo (BA13016).

2.2.2. Raster data structure

Raster layers represent continuous features such as aerial and satellite imagery, as well as Digital Elevation Models (DEM), which store elevation data across a surface (see "Digital elevation model (DEM)" section for more information). The value of each pixel of a thematic raster layer represents the attribute value. For satellite imagery, pixel values represent physical values of surface reflectances and for terrain-model data pixel values represent absolute topographic heights (Campbell, 2002). For example, if the resolution of the image is 25 m, then each 25 m pixel in the image will be representative of the average value of that 25 m on the ground. In Figure 3, a land cover vector layer was transformed to a raster to visualize the pixel differences and value types.

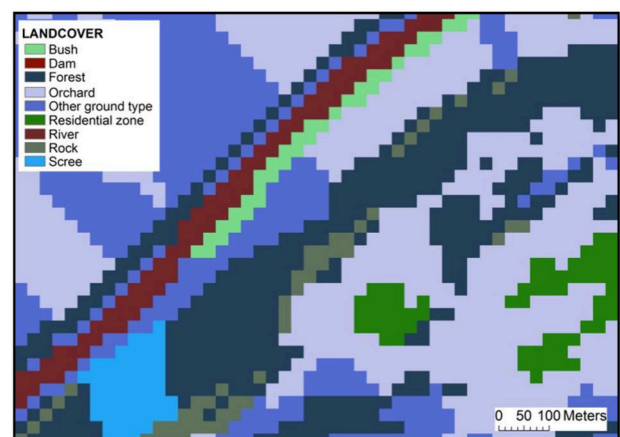


Figure 3. A map displaying the raster data structure of a landcover layer which was converted from vector format. The legend denotes the landcover properties which are displayed. Reproduced with permission of Swisstopo (BA13016).

2.3. GIS functionalities

The main functionalities of GIS include interfaces for database management, geoprocessing and spatial analysis, as well as visualisation and map creation. Here, these functionalities are introduced, and the specific tools will be discussed more in-depth in further sections.

2.3.1. Database management

A GIS database is used to store, create, organize, manipulate, and query spatial and their linked non-spatial datasets (Burrough & Mcdonnell, 1998). GIS facilitates the link between geographic and non-geographic data and enables the comparison and assessment of various data types (Rigaux et al., 2002). Compared to traditional (non-spatial) databases (e.g., those without geographic coordinates), geographic databases make explicit locational distinctions for each dataset stored in the database, that is, all geographic information is linked to a location on the earth's surface through coordinates (Arctur & Zeiler, 2004).

2.3.2. Geoprocessing and spatial analysis

GIS can be used to calculate, analyse, and manipulate spatial data to examine spatial relationships and to create new data (De Smith et al., 2007). Basic manipulations of geographic datasets such as extracting new information from existing layers or clipping specific geographic extents can be referred to as geoprocessing. Geoprocessing refers to the process of creating a new geographic data layer after the calculation of an input layer(s) (Wade & Sommer, 2006).

Spatial analysis is a central concept in GIS and refers to more complex calculations which have been developed from various quantitative methodologies outside of GIS and incorporated into GIS over time (Conolly & Lake, 2006; De Smith et al., 2007; Longley et al., 1999). Spatial analyses can be used to summarize and analyse the spatial properties of geographic distributions, to solve spatial problems through modelling, and finally, to aid in spatial decision making (Longley et al., 1999). Because this is a basic introduction to GIS, we will focus more on the geoprocessing aspects of GIS and touch briefly on more complex spatial analyses.

2.3.3. Data visualisation and map creation

One of the main benefits of GIS is that it gives researchers the ability to visualize their data in a variety of formats, which enables the creation of new links and relationships between spatial entities while discovering spatial interactions that were perhaps unknown before. GIS is also a common cartographic platform which facilitates the production of maps,

either to present a geographic area or the results of data analysis using spatial data. The results of GIS analysis can be displayed as maps using a variety of symbolisations and annotation features as the final step in a GIS investigation.

3. Geographic data

GIS data represent real-world phenomena based on their geometric properties, attributes, and topology (Burrough & Mcdonnell, 1998).

3.1. File types and extensions

Often, GIS software has its own file types for vector and raster data. For example, the Environmental Systems Research Institute's (ESRI) commercial ArcGIS software most commonly uses shapefiles for vector data and grids for raster data. Shapefiles and grids have come to be some of the most commonly known GIS formats and can often be read by and incorporated into other GIS programs without the need for conversion, as is the case for Quantum GIS (QGIS). It is important to note that most layers are made up of multiple files with different file extensions. For example, a shapefile may look like a single file in the GIS, but actually, it is composed of multiple file extensions. It is detrimental to the data layer if one of these extensions gets lost or deleted. Some of the extensions often associated with shapefiles are: *.shp (required file that stores the feature geometry), *.shx (required file that stores the index of the feature geometry), *.dbf (database table that stores the attribute information) or *.prj (file that stores the coordinate system information).

Multiple raster dataset file formats including ASCII (*.asc), GeoTiff (*.tif), JPEG (*.jpg), and GIFs (*.gif), to name only a few, can be supported in most GIS software. Again, it is important to keep all files when copying and moving data. For example, a Geotiff file often comes with two files, *.tif and *.tfw. The *.tif holds the image, and the *.tfw holds the information about the georeference.

3.2. Common GIS data types

The geometric aspect associated with GIS data is what separates geographic databases from other databases (Burrough & Mcdonnell, 1998). Geographic databases are sometimes referred to as geodatabases, which refer to databases specific to geographic data. Until recently, most GIS data were generated and distributed by government or private industries and had to be purchased. Now, there are many sources for downloading free data (Table 2). The amount of available data largely depends on the

Table 2. Names, websites, and types of data available from some of the various free data sources.

Name	Website	Types of data
Aster GDEM	http://asterweb.jpl.nasa.gov/gdem.asp	30 m DEM of the world
CORINE Land Cover Inventory 2000, 2006, 2012, 2018 (European Environment Agency (EEA))	http://www.eea.europa.eu/data-and-maps/data/corine-land-cover-2006-raster-2 https://www.eea.europa.eu/data-and-maps/data/copernicus-land-monitoring-service-corine	Land cover raster data for Europe 100 m and 250 m resolutions
CropScape	https://nassgeodata.gmu.edu/CropScape/	Cropland data for United States
Global Land Cover (MCD12Q1 v006) 2001–2019	https://lpdaac.usgs.gov/products/mcd12q1v006/	Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Type (MCD12Q1) Version 6 provides global land cover at yearly intervals at 500 m resolution
ESRI GIS for Climate	https://climate-arcgis-content.hub.arcgis.com/search	Various data types for climate
European Environment Agency (EEA)	http://www.eea.europa.eu/data-and-maps/data	Land use, economics, natural resources, water, etc.
National Drought Mitigation Center (NDMC) US Drought Monitor	https://droughtmonitor.unl.edu/DmData/GISData.aspx	Shapefiles of drought for United States from 2000 – present
National Land Cover Database (US) (many years)	https://www.mrlc.gov/data	Land cover raster data for the United States at 25 m resolution
National Weather Service and National Oceanic and Atmospheric Administration (NOAA) Advanced Hydrologic Prediction Service	https://water.weather.gov/precip/	Precipitation data for the United States
OpenStreetMap	www.openstreetmap.org	Global or individual country or state map data
OpenStreetMap Data Extracts	http://download.geofabrik.de/	OpenStreetMap data extracts for different places in the world
PRISM Climate Data	https://prism.oregonstate.edu/	Short-term climate patterns Long-term climate patterns Various data types, by country
United Nations Geographic Information Working Group (UNGIWG) Second Administrative Level Boundaries (SALB)	https://www.unsalb.org/data	
United States Department of Agriculture (USDA) National Agriculture Statistics Survey	https://www.nass.usda.gov/Surveys/Guide_to_NASS_Surveys/Bee_and_Honey/	Pesticide and crop data for United States
United States Geological Survey (USGS) Earth Resources Observation and Science (EROS) Center	https://earthexplorer.usgs.gov/	Aerial Photography Satellite Imagery Elevation Land Cover Digitized Maps Image Gallery Collections
USGS Pesticide National Synthesis Project	https://water.usgs.gov/nawqa/pnsp/usage/maps/county-level/	Estimated annual agricultural pesticide use at county and states level since 1992

area of interest, as urban areas tend to have more data available than rural or remote areas. Another important aspect of geographic data is the meta-data. Metadata are the information provided about the data, thus, data about data (Burrough & McDonnell, 1998). When downloading data online or receiving data from an outside source, metadata are important for obtaining knowledge about when the data were created, for what purpose, by whom, and using which coordinate system, among other things. Without this information, data would be of limited use only. In the following section we will introduce some of the common file formats and extensions, as well as some geographic data examples. The possibilities of types of data in GIS are becoming broader as the technology develops. The following is a non-exhaustive list that shows some of the data types you may encounter when using GIS.

3.2.1. Maps

Topographic maps (Figure 4) are representations of the terrain which include both natural and human-made features, including, relief (in the form of contour lines), towns, villages, roads, and other geographic characteristics (Burrough & McDonnell, 1998).



Figure 4. A sample topographic map (scale 1: 200 000) covering the study area of Fribourg, Switzerland. This map comes from Swisstopo (2013) and must only be used for visualization purposes in this paper. Reproduced with permission of Swisstopo (BA13016).

They are those maps which people are most familiar with. In GIS, topographic maps are stored in raster format in their entirety but can also be broken down



Figure 5. A sample orthophoto, with a resolution of 50 cm, showing the study area of Fribourg, Switzerland. This photo comes from Swisstopo (2013) and must only be used for visualization purposes in this paper. Reproduced with permission of Swisstopo (BA13016).

into their respective categories (rivers, buildings, roads, etc.) in vector layers.

3.2.2. Aerial imagery

Photos taken from the air, usually from an aircraft, are referred to as aerial imagery or photographs (Figure 5). These images provide valuable information about landscape changes by comparing photos over time. When aerial images are georeferenced, i.e., located with respect to a geographic or projected coordinate system, they can be used as raster layers in GIS (Burrough & McDonnell, 1998). The different types of aerial images include black and white, true colour, panchromatic, and infrared and can each be used for different purposes (Campbell, 2002).

3.2.3. Satellite imagery

Satellite images are those that have been taken from satellites. Landsat (Figure 6) is one of the longest running platforms for image acquisition using the Thematic Mapper (TM) or the Enhanced Thematic Mapper (ETM) satellite image acquisition systems (<http://landsat.gsfc.nasa.gov/>) which provide multi-spectral imagery for the whole world (Campbell, 2002). Satellite imagery is often expensive to buy as the equipment is difficult to operate and maintain, but conveniently, Landsat and other satellite images can be added as raster layers directly into the GIS interface.

3.2.4. Digital Elevation Model (DEM)

A DEM (Figure 7) is a raster dataset that represents the altitude of the terrain. Depending on the resolution of the image, each pixel represents the



Figure 6. A section of the Swisstopo (2013) Landsat image with a resolution of 25 m, covering the study area of Fribourg, Switzerland. Reproduced with permission of Swisstopo (BA13016).

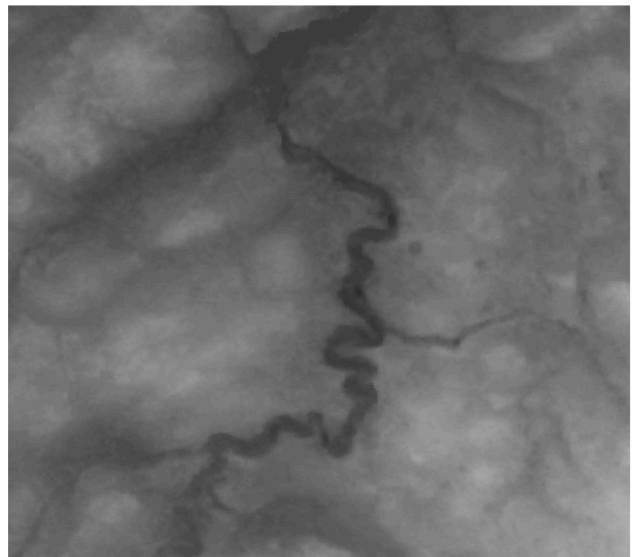


Figure 7. A sample digital elevation model (DEM) (30 m resolution) covering the study area of Fribourg, Switzerland. This map was a free download from ASTER GDEM which was projected to the Swiss projected coordinate system (CH1903 LV03).

average altitude of the terrain covered by that pixel. For example, if the image resolution is 30 m, the altitude of the terrain on the ground that is covered by that pixel within that 30 m square is averaged and the value is assigned to that pixel.

With a DEM, numerous calculations can be performed including the calculation of slope (in degrees or percent) of the terrain, the direction each slope is facing (aspect), and where shadows exist when the sun is hitting the terrain from a certain angle (hill-shade). A DEM is also the base dataset for calculations of solar radiation, hydrographic, and various modelling calculations.

3.2.5. Thematic vector data

Vector (Figure 2) data represent a broad range of information, as they include all layers represented by points, lines, or polygons. In a GIS database, common vector layers might represent administrative boundaries (the locations of countries, provinces, states, cantons, etc.), hydrography (rivers, dams, lakes, etc.), built-up areas (buildings, stations, airports, etc.), and land cover.

4. Coordinate systems

This section will briefly introduce the concept of coordinate systems in GIS. For any analysis, it is important to know the coordinate system of the data which represent the study area (region of interest) and to ensure that all of your data layers are already in, or can be converted to, that coordinate system. Basic knowledge about coordinate systems is recommended before any type of spatial analysis is conducted as they can affect the resulting outputs. It is also important to know the capabilities and weaknesses of each specific coordinate system used in analysis. For more in-depth information about coordinate systems in GIS see Seeger (1999).

4.1. Geographic coordinate systems

A geographic coordinate system is defined by the ellipsoid and datum that are used to calculate longitudes and latitudes which represent locations on the Earth's surface (Snyder, 1987). It is represented in degrees, minutes, and seconds north or south of the equator (latitude) or east and west of the equator (longitude) (i.e., Fribourg, Switzerland is at 46° 48'

00" N, 7° 09' 00" E), and is the simplest solution for representing places on the globe (Burrough & McDonnell, 1998). Decimal degrees are also used to represent latitude/longitude coordinates and can be calculated with the following equation: $\text{Decimal Degrees} = \text{Degrees} + \text{Minutes}/60 + \text{Seconds}/3,600$ (<http://support.esri.com/en/knowledgebase/techar-ticles/detail/27215>) (i.e., same location above would be represented as 46.8 N, 7.15 E). There are various online sources for calculating conversions between latitude/longitude and decimal degrees (e.g., <http://transition.fcc.gov/mb/audio/bickel/DDMMSS-decimal.html>). The most commonly used geographic coordinate system is the World Geodetic System (WGS) 1984 (Figure 8). However, geographic coordinate systems do not transfer nicely into two dimensions, causing major distortions in distance, area, shape, or direction (Snyder, 1987; Seeger, 1999) on a map or computer screen. To solve these issues, projected coordinate systems are used.

4.2. Projected coordinate systems

Projected coordinate systems are created to allow a two-dimensional representation on a screen or map sheet (Snyder, 1987). Latitudes and longitudes are converted to meters with respect to the centre of projection by applying calculations to geographic coordinate systems to counteract and offset the distortions on the map. Map projections have been developed for both local and regional scales. There is a large body of research based on the calculations of map projections (see Snyder, 1987 or Seeger, 1999 for more information). One of the most important things to remember is that all map projections

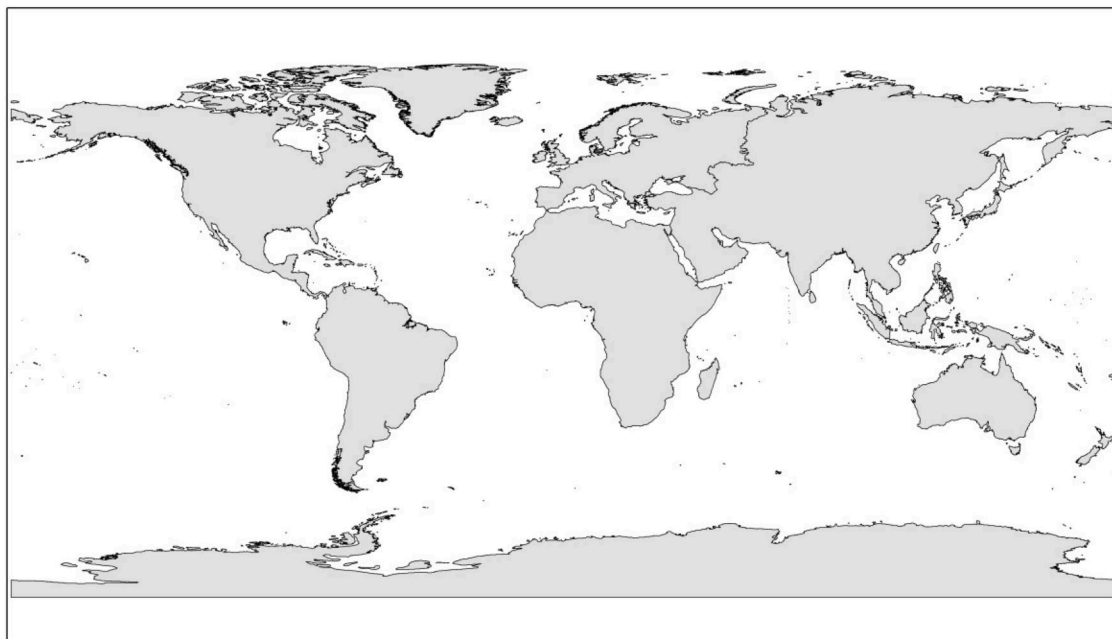


Figure 8. A map of the world shown in the WGS 84 geographic coordinate system.